

Sparse Continuation-Based Eigenvalue Tracking for Power System DDAEs[★]

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ABSTRACT

In this paper, we formulate a continuation method for tracking eigenvalue trajectories in power system models with time-delayed measurement and control signals. Such delays are known to weaken damping and reduce stability margins if not properly accounted for in stability analysis and control design. The proposed formulation follows eigenpairs directly from sparse delay differential-algebraic equation (DDAE) models with one or multiple delayed variables, while allowing the continuation parameter to represent system parameters, delay magnitudes, or parameters associated with non-constant wide area measurement system (WAMS) delays. The proposed method is validated on a modified IEEE 39-bus system and on a real-world-scale dynamic model of the Irish transmission network, demonstrating accurate tracking of critical eigenvalue trajectories and clear computational advantages over repeated eigensolution-based analysis.

1. Introduction

1.1. Motivation

Time delays in power system control loops arise from measurement, data processing, and communication. These delays, which typically range from a few milliseconds to hundreds of milliseconds, can lead to sustained oscillations and compromise stability [1–5]. These effects can be assessed through small-signal stability analysis (SSSA), where eigenvalue analysis plays a central role [6]. However, in the presence of delays, eigenvalue analysis typically relies on spectral approximations, making repeated eigenvalue computations expensive and tracking eigenvalue trajectories across parameter variations nontrivial, particularly in large-scale settings. Addressing the problem of tracking eigenvalue trajectories in power system models with time delays is the main goal of this paper.

1.2. Literature Review

Power system dynamics in short-term stability analysis are commonly modeled through a set of differential-algebraic equations (DAEs) [7]. Introducing time delays transforms the model into a set of delay differential-algebraic equations (DDAEs). In SSSA, the stability of linearized DAEs and DDAEs is determined by their eigenvalues, defined as the roots of the associated characteristic equation. For DDAEs, this equation is transcendental and gives rise to an infinite spectrum [6]. Approximation techniques such as Padé polynomials or spectral discretization yield a finite-dimensional linear eigenvalue problem, which can then be

solved using standard numerical algorithms [8]. These include dense-matrix methods, such as QR or QZ, which are suitable for small to medium-sized systems but quickly become computationally prohibitive as the problem scales; and sparse techniques, such as Krylov methods, which can obtain selected parts of the spectrum in large systems, but rely on user-defined spectral transformations and target regions. This makes their repeated use nontrivial when system parameters vary and eigenvalue trajectories must be followed.

This difficulty motivates methods that follow eigenvalues as continuous functions of the varying parameters, rather than recomputing and re-identifying them independently at each parameter value. Such a framework is provided by continuation methods. The core idea is to incrementally update the solution of a parameter-dependent problem using previously computed values. In the context of eigenvalue tracking, this solution consists of a subset of the system's eigenpairs, i.e., eigenvalues with their associated eigenvectors. For systems without delays, [9] proposes a continuation-based algorithm that employs Newton's method to follow eigenvalue trajectories, highlighting the benefit of including a predictor step before each iteration. The authors in [10] combine this approach with a Cayley transform that updates the dimension of the considered subspace at every iteration, focusing the computation on the modes most critical for stability. To evaluate power system stability margins, [11] derives eigenvalue and eigenvector sensitivities with respect to varying parameters and traces their trajectories through numerical integration. In all these works, time delays are neglected.

Among the few works that do consider delays, existing formulations are limited to constant delay models [12, 13]. Moreover, the formulation in [12] is based on dense Sylvester-type equations, limiting applicability to small systems. On the other hand, the formulation in [13] is limited to a specific choice of the continuation parameter, i.e., the delay magnitude [13]. Addressing these limitations is the focus of this paper.

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1.3. Contribution

Motivated by the limitations identified above, this paper develops a continuation-based eigenvalue tracking formulation for large-scale power system DDAEs. The main advances compared with existing techniques are as follows:

- The formulation works directly with the sparse DDAE eigenvalue problem. This avoids dense matrix equations, whose computational cost becomes prohibitive as the problem scales.
- The continuation parameter is not limited to the delay magnitude. It can represent system parameters, control parameters, delay magnitudes, or combinations of them, allowing different stability studies to be treated within the same continuation setting.
- The formulation can handle multiple delayed variables and can account for non-constant wide area measurement system (WAMS) delays, including stochastic latency effects and data packet dropouts. This supports a more complete assessment of wide-area controllers.

1.4. Paper Organization

The remainder of the paper is organized as follows. Section 2 introduces preliminaries on SSSA. Section 3 discusses the formulation and implementation aspects of the proposed eigenvalue tracking approach. Section 4 presents simulation results, based on the modified 39-bus system and a real-world scale model of the Irish transmission system, demonstrating the accuracy and computational performance of the method. Finally, Section 5 draws conclusions and outlines directions for future work.

2. Small-Signal Stability Analysis

2.1. DDAE Power System Model

Short-term power system stability in the presence of delays can be studied through a set of nonlinear DDAEs in semi-implicit¹ form [6]

$$\begin{bmatrix} T & \mathbf{0}_{n,m} \\ R & \mathbf{0}_{m,m} \end{bmatrix} \begin{bmatrix} \dot{\mathbf{x}}(t) \\ \dot{\mathbf{y}}(t) \end{bmatrix} = \begin{bmatrix} \mathbf{f}(\mathbf{x}, \mathbf{y}, \mathbf{x}_d, \mathbf{y}_d) \\ \mathbf{g}(\mathbf{x}, \mathbf{y}, \mathbf{x}_d, \mathbf{y}_d) \end{bmatrix} \quad (1)$$

where $\mathbf{x} = \mathbf{x}(t) : [0, \infty) \rightarrow \mathbb{R}^n$ are the states of dynamic devices such as generators, dynamic loads, and controllers; $\mathbf{y} = \mathbf{y}(t) : [0, \infty) \rightarrow \mathbb{R}^m$ are algebraic variables associated with network equations and auxiliary control setpoints; $\mathbf{f}$ and $\mathbf{g}$ are nonlinear functions; $T \in \mathbb{R}^{n \times n}$, $R \in \mathbb{R}^{m \times n}$; and $\mathbf{0}_{n,m}$ denotes the $n \times m$ zero matrix. The delayed state and algebraic variables are denoted as $\mathbf{x}_d$ and $\mathbf{y}_d$, with:

$$\begin{aligned} \mathbf{x}_d(t) &= \{\mathbf{x}(t - \tau_1), \mathbf{x}(t - \tau_2), \dots, \mathbf{x}(t - \tau_\mu)\} \\ \mathbf{y}_d(t) &= \{\mathbf{y}(t - \tau_1), \mathbf{y}(t - \tau_2), \dots, \mathbf{y}(t - \tau_\mu)\} \end{aligned} \quad (2)$$

¹The semi-implicit form is often adopted because it can offer certain practical advantages, such as increased sparsity as well as flexibility in handling state and algebraic variables, see [6].

where $\tau_j > 0$, $j = 1, 2, \dots, \mu$, denotes the j -th delay and μ is the total number of delays. For brevity, we denote the j -th delayed variable as:

$$\begin{aligned} \mathbf{x}_d^j &= \mathbf{x}(t - \tau_j), \\ \mathbf{y}_d^j &= \mathbf{y}(t - \tau_j), \end{aligned} \quad (3)$$

where $j = 1, 2, \dots, \mu$.

2.2. Eigenvalue Analysis

To study how the system responds to parameter variations, we introduce a scalar continuation parameter $p \in \mathbb{R}$, used to parametrize selected system quantities of interest. Then, (1) is rewritten as follows:

$$\begin{bmatrix} T(p) & \mathbf{0}_{n,m} \\ R(p) & \mathbf{0}_{m,m} \end{bmatrix} \begin{bmatrix} \dot{\mathbf{x}} \\ \dot{\mathbf{y}} \end{bmatrix} = \begin{bmatrix} \mathbf{f}(\mathbf{x}, \mathbf{y}, \mathbf{x}_d, \mathbf{y}_d, p) \\ \mathbf{g}(\mathbf{x}, \mathbf{y}, \mathbf{x}_d, \mathbf{y}_d, p) \end{bmatrix} \quad (4)$$

An equilibrium $(\mathbf{x}_o, \mathbf{y}_o) := [\mathbf{x}_o^\top, \mathbf{y}_o^\top]^\top$ ($\top$ indicating the transpose) of (4) can be defined, assuming the system has remained at rest for at least the maximum delay. Considering small disturbances, (4) can be linearized around the equilibrium, as follows:

$$\begin{aligned} \begin{bmatrix} T(p) & \mathbf{0}_{n,m} \\ R(p) & \mathbf{0}_{m,m} \end{bmatrix} \begin{bmatrix} \Delta \dot{\mathbf{x}} \\ \Delta \dot{\mathbf{y}} \end{bmatrix} &= \begin{bmatrix} \mathbf{f}_x(p) & \mathbf{f}_y(p) \\ \mathbf{g}_x(p) & \mathbf{g}_y(p) \end{bmatrix} \begin{bmatrix} \Delta \mathbf{x} \\ \Delta \mathbf{y} \end{bmatrix} \\ &+ \sum_{j=1}^{\mu} \left(\begin{bmatrix} \mathbf{f}_{x,j}(p) & \mathbf{f}_{y,j}(p) \\ \mathbf{g}_{x,j}(p) & \mathbf{g}_{y,j}(p) \end{bmatrix} \begin{bmatrix} \Delta \mathbf{x}_d^j \\ \Delta \mathbf{y}_d^j \end{bmatrix} \right) \end{aligned} \quad (5)$$

where $\Delta \mathbf{x} = \mathbf{x} - \mathbf{x}_o$, $\Delta \mathbf{y} = \mathbf{y} - \mathbf{y}_o$; $\mathbf{f}_x$, $\mathbf{f}_y$, $\mathbf{g}_x$, $\mathbf{g}_y$ and $\mathbf{f}_{x,j}$, $\mathbf{f}_{y,j}$, $\mathbf{g}_{x,j}$, $\mathbf{g}_{y,j}$ are the delay-free and delayed Jacobians, respectively, evaluated at $(\mathbf{x}_o, \mathbf{y}_o)$. Equation (5) is of the form:

$$\mathbf{E}(p) \dot{\mathbf{x}} = \mathbf{A}_0(p) \mathbf{x} + \sum_{j=1}^{\mu} \mathbf{A}_j(p) \mathbf{x}_d^j \quad (6)$$

where

$$\begin{aligned} \mathbf{x} &= (\Delta \mathbf{x}, \Delta \mathbf{y}), \\ \mathbf{x}_d^j &= (\Delta \mathbf{x}_d^j, \Delta \mathbf{y}_d^j), \end{aligned}$$

and

$$\begin{aligned} \mathbf{E}(p) &\equiv \begin{bmatrix} T(p) & \mathbf{0}_{n,m} \\ R(p) & \mathbf{0}_{m,m} \end{bmatrix}, \quad \mathbf{A}_0(p) \equiv \begin{bmatrix} \mathbf{f}_x(p) & \mathbf{f}_y(p) \\ \mathbf{g}_x(p) & \mathbf{g}_y(p) \end{bmatrix} \\ \mathbf{A}_j(p) &\equiv \begin{bmatrix} \mathbf{f}_{x,j}(p) & \mathbf{f}_{y,j}(p) \\ \mathbf{g}_{x,j}(p) & \mathbf{g}_{y,j}(p) \end{bmatrix} \end{aligned} \quad (7)$$

Applying the Laplace transform to (6):

$$(s\mathbf{E}(p) - \mathbf{A}_0(p) - \sum_{j=1}^{\mu} \mathbf{A}_j(p)e^{-s\tau_j}) \mathcal{L}\{\mathbf{x}\} = \mathbf{E}(p)\mathbf{x}(0) \quad (8)$$

where s is a complex variable in the S -plane. The associated eigenvalue problem is

$$\mathbf{P}(s, p) \boldsymbol{\phi} = \mathbf{0}_{r,1}, \quad r = n + m \quad (9)$$

where any value of s that satisfies (9) is an eigenvalue of the matrix function:

$$\mathbf{P}(s, p) = s\mathbf{E}(p) - \mathbf{A}_0(p) - \sum_{j=1}^{\mu} \mathbf{A}_j(p)e^{-s\tau_j} \quad (10)$$

with ϕ being the associated right eigenvector. The matrix function $\mathbf{P}(s, p)$ plays an important role in the stability and dynamic behavior of (6).

3. Eigenvalue tracking

This section presents a continuation-based approach for tracking the eigenvalues of power systems with time delays. The method is first introduced for a single delayed variable, then generalized to systems with multiple delays, and extended to allow the delay magnitude itself to vary as the continuation parameter. The formulation is also extended to account for realistic time-varying communication delays, using a wide area measurement system (WAMS) delay model that accounts for stochasticity and data packet dropouts.

3.1. Single Delay

We begin, for simplicity, by considering a single delay $\tau = \tau_1$ ($\mu = 1$). Equation (10) takes the form:

$$\mathbf{P} = s\mathbf{E} - \mathbf{A}_0 - \mathbf{A}_1 e^{-s\tau} \quad (11)$$

where the dependence on p is omitted for brevity. Differentiation of (9) with respect to p gives:

$$\mathbf{P}' \phi + \mathbf{P} \phi' = \mathbf{0}_{r,1} \quad (12)$$

where

$$\mathbf{P}' = s'\mathbf{E} + s\mathbf{E}' - \mathbf{A}'_0 - \mathbf{A}'_1 e^{-s\tau} + \tau s' \mathbf{A}_1 e^{-s\tau} \quad (13)$$

with:

$$\begin{aligned} s' &= \partial s / \partial p, & \mathbf{E}' &= \partial \mathbf{E} / \partial p, & \mathbf{A}'_0 &= \partial \mathbf{A}_0 / \partial p, \\ \mathbf{A}'_1 &= \partial \mathbf{A}_1 / \partial p, & \phi' &= \partial \phi / \partial p. \end{aligned}$$

Equivalently, equation (12) is:

$$(s\mathbf{E} - \mathbf{A}_0 - \mathbf{A}_1 e^{-s\tau})\phi' + (\mathbf{E} + \tau \mathbf{A}_1 e^{-s\tau})\phi s' = - (s\mathbf{E}' - \mathbf{A}'_0 - \mathbf{A}'_1 e^{-s\tau})\phi \quad (14)$$

System (14), which describes the evolution of a single eigenpair with respect to p , consists of r equations and $r + 1$ unknowns, s and ϕ . To make this system well-posed, it is closed by imposing the following eigenvector normalization:

$$\phi^\top \phi = c \quad (15)$$

where c is an imposed constant, e.g., $c = 1$. Differentiation of (15) gives:

$$\phi^\top \phi' = 0 \quad (16)$$

Combining (14) and (16):

$$\begin{bmatrix} s\mathbf{E} - \mathbf{A}_0 - \mathbf{A}_1 e^{-s\tau} & (\mathbf{E} + \tau \mathbf{A}_1 e^{-s\tau})\phi \\ \phi^\top & 0 \end{bmatrix} \begin{bmatrix} \phi' \\ s' \end{bmatrix} = \begin{bmatrix} - (s\mathbf{E}' - \mathbf{A}'_0 - \mathbf{A}'_1 e^{-s\tau})\phi \\ 0 \end{bmatrix} \quad (17)$$

Splitting real and imaginary parts, i.e.

$$\begin{aligned} \phi &= \phi_r + j\phi_i, \\ s &= s_r + js_i, \end{aligned}$$

and by setting $\mathbf{A}_1 e^{-s_r \tau} \cos(s_i \tau) = \mathbf{C}$, $\mathbf{A}_1 e^{-s_r \tau} \sin(s_i \tau) = \mathbf{S}$, $\mathbf{A}'_1 e^{-s_r \tau} \cos(s_i \tau) = \mathbf{C}_D$ and $\mathbf{A}'_1 e^{-s_r \tau} \sin(s_i \tau) = \mathbf{S}_D$ we arrive at the following system:

$$\mathbf{M}(\mathbf{y}) \mathbf{y}' = \mathbf{h}(\mathbf{y}) \quad (18)$$

where $\mathbf{y} = \mathbf{y}(p) = (\phi_r, \phi_i, s_r, s_i)$, with $\mathbf{y} \in \mathbb{R}^{2r+2}$; and:

$$\mathbf{M}(\mathbf{y}) = \begin{bmatrix} \mathbf{M}_1 & \mathbf{M}_2 \\ \mathbf{M}_3 & \mathbf{0}_{2,2} \end{bmatrix}$$

The quantities $\mathbf{h}$, $\mathbf{M}_1$, $\mathbf{M}_2$, $\mathbf{M}_3$ are defined as:

$$\begin{aligned} \mathbf{h} &= \begin{bmatrix} (-s_r \mathbf{E}' + \mathbf{A}'_0 + \mathbf{C}_D)\phi_r + (s_i \mathbf{E}' + \mathbf{S}_D)\phi_i \\ -(s_i \mathbf{E}' + \mathbf{S}_D)\phi_r + (-s_r \mathbf{E}' + \mathbf{A}'_0 + \mathbf{C}_D)\phi_i \\ 0 \\ 0 \end{bmatrix} \\ \mathbf{M}_1 &= \begin{bmatrix} s_r \mathbf{E} - \mathbf{A}_0 - \mathbf{C} & -s_i \mathbf{E} - \mathbf{S} \\ s_i \mathbf{E} + \mathbf{S} & s_r \mathbf{E} - \mathbf{A}_0 - \mathbf{C} \end{bmatrix} \\ \mathbf{M}_2 &= \begin{bmatrix} \mathbf{E}\phi_r + \tau(\mathbf{C}\phi_r + \mathbf{S}\phi_i) & -\mathbf{E}\phi_i - \tau(\mathbf{C}\phi_r - \mathbf{S}\phi_i) \\ \mathbf{E}\phi_i + \tau(\mathbf{C}\phi_r - \mathbf{S}\phi_i) & \mathbf{E}\phi_r + \tau(\mathbf{C}\phi_r + \mathbf{S}\phi_i) \end{bmatrix} \\ \mathbf{M}_3 &= \begin{bmatrix} \phi_r^\top & -\phi_i^\top \\ \phi_i^\top & \phi_r^\top \end{bmatrix} \end{aligned}$$

3.2. Generalization to Multiple Delays

When multiple delays are present in the system, the expression (12) holds, where in this case the matrix function $\mathbf{P}$ is given by (10). Differentiating (10) with respect to the continuation parameter p gives:

$$\mathbf{P}' = s'\mathbf{E} + s\mathbf{E}' - \mathbf{A}'_0 - \sum_{j=1}^{\mu} (\mathbf{A}'_j - \tau_j s' \mathbf{A}_j) e^{-s\tau_j} \quad (19)$$

Substituting into (12) yields:

$$\begin{aligned} (\mathbf{E} + \sum_{j=1}^{\mu} \tau_j \mathbf{A}_j e^{-s\tau_j})\phi s' + (s\mathbf{E} - \mathbf{A}_0 - \sum_{j=1}^{\mu} \mathbf{A}_j e^{-s\tau_j})\phi' \\ = - (s\mathbf{E}' - \mathbf{A}'_0 - \sum_{j=1}^{\mu} \mathbf{A}'_j e^{-s\tau_j})\phi \end{aligned} \quad (20)$$

Combining with (16) and splitting eigenvalue and eigenvector into real and imaginary parts leads to a system in the form of (18), where $\mathbf{M}$ and $\mathbf{h}$ are defined to account for all μ delays. Their detailed definitions are provided in the Appendix.

3.3. Delay Magnitude as Varying Parameter

In this subsection, we track the evolution of eigenvalues as the delay magnitude is varied. We first consider the case where the system contains a single delayed variable, whose time delay has a varying magnitude and is denoted by τ_ℓ . The corresponding delayed state matrix is denoted by $\mathbf{A}_\ell$. Given $p = \tau_\ell$, the matrix function $\mathbf{P}$ is:

$$\mathbf{P} = s\mathbf{E} - \mathbf{A}_0 - \mathbf{A}_\ell e^{-sp} \quad (21)$$

Matrices $\mathbf{E}$, $\mathbf{A}_0$ and $\mathbf{A}_\ell$ do not explicitly depend on τ , hence differentiation of (21) with respect to p yields:

$$\mathbf{P}' = s'\mathbf{E} + \mathbf{A}_\ell(s'p + s)e^{-sp}$$

Substituting into (12) yields:

$$(s\mathbf{E} - \mathbf{A}_0 - \mathbf{A}_\ell e^{-sp})\phi' + (\mathbf{E} + \mathbf{A}_\ell p e^{-sp})\phi s' = -\mathbf{A}_\ell e^{-sp} s\phi \quad (22)$$

By splitting real and imaginary parts of (22), we arrive at a system in the form of (18), where $\mathbf{M}$ and $\mathbf{h}$ are in this case functions of the varying delay magnitude $p = \tau_\ell$. Their definitions are detailed in the Appendix.

We extend the above formulation to systems with multiple delays, treating one of them as the continuation parameter, i.e., $\tau_\ell = p$. The definition of $\mathbf{P}$ in this case is as follows:

$$\mathbf{P} = s\mathbf{E} - \mathbf{A}_0 - \sum_{j=1}^{\mu} \mathbf{A}_j e^{-s\tau_j} - \mathbf{A}_\ell e^{-sp} \quad (23)$$

We note that the term corresponding to τ_ℓ is kept separate, even if τ_ℓ equals some τ_j , $j = 1, 2, \dots, \mu$. Given that the matrices $\mathbf{E}$, $\mathbf{A}_0$, $\mathbf{A}_\ell$ and $\mathbf{A}_j$ do not explicitly depend on τ_ℓ , the derivative of (23) with respect to p is:

$$\mathbf{P}' = s'\mathbf{E} + \sum_{j=1}^{\mu} \tau_j s' \mathbf{A}_j e^{-s\tau_j} + \mathbf{A}_\ell(s'p + s)e^{-sp} \quad (24)$$

and (12) is equivalently written as:

$$(s\mathbf{E} - \mathbf{A}_0 - \mathbf{A}_\ell e^{-sp} - \sum_{j=1}^{\mu} \mathbf{A}_j e^{-s\tau_j})\phi' + (\mathbf{E} + \mathbf{A}_\ell p e^{-sp} + \sum_{j=1}^{\mu} \tau_j \mathbf{A}_j e^{-s\tau_j})\phi s' = -\mathbf{A}_\ell e^{-sp} s\phi \quad (25)$$

By splitting the eigenvalue and eigenvector into real and imaginary parts we arrive at a system in the form of (18). The derivation of $\mathbf{M}$ and $\mathbf{h}$ is provided in the Appendix.

3.4. Communication Delays with Noise and Data Dropouts

The derivations above consider constant delays. In this section, we show how they can be generalized to capture the effects of realistic WAMS latencies with noise and data packet dropouts. To this end, we consider the composite

time-varying delay model proposed in [1], in which, for the purpose of SSSA, packet dropouts and noise are represented through the functions $h_p(s)$ and $h_s(s)$, respectively:

$$h_p(s) = \frac{1 - p_{dr}}{sT} [1 + (p_{dr} - 1) \frac{e^{-sT}}{1 - p_{dr}e^{-sT}}] \quad (26)$$

$$h_s(s) = (1 + \frac{\alpha}{1 - p_{dr}} s)^{-b} \quad (27)$$

where α and b are the scale and shape factors of the *Gamma distribution*; p_{dr} is the packet dropout rate; and T is the normal delivery period for each data packet.

Equation (10) takes the adjusted form:

$$\mathbf{P} = s\mathbf{E} - \mathbf{A}_0 - h_p(s)h_s(s)\mathbf{A}_1 e^{-s\tau_0} \quad (28)$$

where τ_0 is the constant component of the WAMS delay. By setting $S_T = h_p(s)h_s(s)e^{-s\tau_0}$, the derivative of (28) with respect to p is:

$$\mathbf{P}' = s'\mathbf{E} + s\mathbf{E}' - \mathbf{A}'_0 - S'_T \mathbf{A}_1 - S_T \mathbf{A}'_1 \quad (29)$$

where

$$\begin{aligned} S'_T &= (\frac{\partial h_p}{\partial s} h_s + h_p \frac{\partial h_s}{\partial s} - \tau_0 h_p h_s) e^{-s\tau_0} s' \\ &= S_{TD} s' \end{aligned} \quad (30)$$

We note that for a constant delay, i.e., $\tau(t) = \tau_0$, the model reduces to $h_p(s) = h_s(s) = 1$ and $\partial h_p / \partial s = \partial h_s / \partial s = 0$; therefore, equations (28) and (29) are simplified to (11) and (13) respectively.

Substituting (28) and (29) into (12) yields:

$$(s\mathbf{E} - \mathbf{A}_0 - S_T \mathbf{A}_1)\phi' + (\mathbf{E} - S_{TD} \mathbf{A}_1)\phi s' = -(\mathbf{E}' - \mathbf{A}'_0 - S_T \mathbf{A}'_1)\phi \quad (31)$$

By splitting real and imaginary parts of (31), we arrive at a system in the form of (18), where the definitions of $\mathbf{M}$ and $\mathbf{h}$ are detailed in the Appendix.

3.5. Numerical Integration

Given a parameter range $[p_{init}, p_{fin}]$, tracking is performed by numerically integrating the parameter-space system (18). For the initial value p_{init} , the state vector $\mathbf{y}(p_{init})$ is obtained by solving the eigenvalue problem (9). An effective method, in our experience, is to transform the linearized DDAEs into an equivalent system of partial differential equations, and then reduce it through Chebyshev polynomials to a finite-dimensional linear eigenvalue problem. The spectral discretization technique used in this paper is described in [14]. In subsequent steps, if the examined variation alters the power flow solution, the latter is recomputed. Otherwise, if the varying parameters correspond to dynamic devices, the last step is skipped; the operating point is updated from the steady-state DDAEs and the matrices $\mathbf{A}_0$, $\mathbf{A}_1$ and $\mathbf{E}$ are reconstructed. Matrix derivatives are computed numerically using first-order finite differences.

Integration proceeds iteratively until $p = p_{fin}$; firstly, with step size Δp , the parameter is updated as $p_{k+1} =$

$p_k + \Delta p$. Then, the mass matrix $\mathbf{M}(\mathbf{y}_k)$ and right-hand side $\mathbf{h}(\mathbf{y}_k)$ are evaluated, and the state vector is advanced using a numerical integration method. For instance, using the forward euler method (FEM) gives:

$$\mathbf{y}_{k+1} = \mathbf{y}_k + \Delta p \mathbf{M}^{-1}(\mathbf{y}_k) \mathbf{h}(\mathbf{y}_k) \quad (32)$$

The product $\mathbf{M}^{-1}(\mathbf{y}_k) \mathbf{h}(\mathbf{y}_k)$ can be determined through LU decomposition of $\mathbf{M}(\mathbf{y}_k)$.

The choice of integration method for (18) involves a trade-off between speed and accuracy; for example, the trapezoidal method (TM) is more accurate than FEM, but leads to additional computational cost. We note that regardless of the choice, numerical integration schemes inherently introduce discretization error, which depends on the step size. This error can be effectively mitigated by adding a Newton-based corrector step that recomputes the solution within a prescribed tolerance, at additional computational cost. Combining equations (9) and (15) for $c = 1$, the Newton-based corrector step is formulated as follows:

$$\begin{bmatrix} \frac{\partial \mathbf{P}}{\partial \phi} & \frac{\partial \mathbf{P}}{\partial s} \\ \hat{\phi}^\top & 0 \end{bmatrix} \begin{bmatrix} \Delta \phi \\ \Delta s \end{bmatrix} = - \begin{bmatrix} \mathbf{P}(s^{[j]}, \phi^{[j]}) \\ \hat{\phi}^\top \phi^{[j]} - 1 \end{bmatrix}$$

For the results presented in Section 4, unless stated otherwise, a corrector sub-routine with an error tolerance of 10^{-8} is applied after each step.

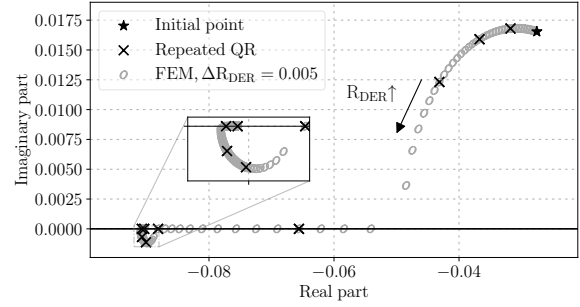
4. Case Study

In this section, we apply the continuation-based eigenvalue tracking formulation of Section 3 to two power system models. First, we consider a modified version of the IEEE 39-bus system [15], and then a real-world scale model of the Irish transmission system. All simulations are carried out with the power system software tool Dome [16]. The version of the software employed in this paper relies on LAPACK 3.12.0 for QR factorization and KLU 2.3.6 for LU factorization. For the sparse Krylov-Schur (KS) method it relies on SLEPc 3.24.1 [17].

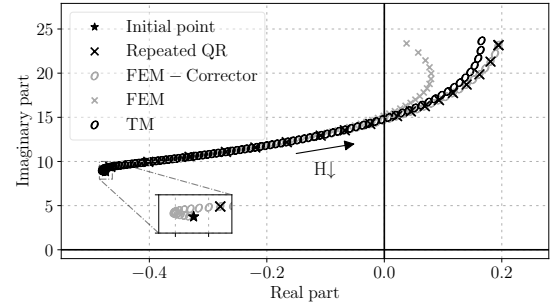
4.1. 39-Bus System

We consider a modified version of the IEEE 39-bus system, wherein the machines at buses 30, 34, 35, 36, 37 are replaced by distributed energy resources (DERs) of equal capacity. Each DER is modeled with an inner current control loop and outer voltage and frequency regulation loops [18].

We begin by demonstrating the proposed tracking method on the system's frequency regulation (FR) mode, also referred to as common low-frequency mode in the literature [19, 20]. This is a system-wide mode originating from the dynamics of frequency control loops [21]. For the examined system, the natural frequency of the FR mode is initially 0.005 Hz. To illustrate the process, the frequency control droop constant R_{DER} of the five DERs is varied from 0.05 to 0.5. A single delay of $\tau = 0.01$ s is introduced in the frequency signal of the power system stabilizer (PSS) connected to the synchronous generator (SG) at bus 31. The



(a) FR mode: increasing the DER frequency controllers' droop constant.



(b) Critical electromechanical mode: Reducing the inertia of the SG at bus 39.

Figure 1: 39-bus system: tracking in the presence of a single delay.

traced eigenvalue, initially at $-0.0276 + j0.0165$, follows the trajectory shown in Fig. 1a. Tracking based on (18) accurately captures the trajectory and shows how the oscillatory behavior of the FR mode is progressively suppressed as the relative share of frequency control shifts from the DERs to the SGs. At the point where the complex pair coalesces on the real axis, the eigenvalue becomes defective with algebraic multiplicity 2 and geometric multiplicity 1. This corresponds to a simple quadratic fold [21, 22]. As R_{DER} is further increased, the defective eigenvalue splits into two distinct real eigenvalues. The tracking method inherently follows one of the two emerging branches, as determined by the evolution of (18). In this example, the eigenvalue on the left branch of the fold is traced, and Fig. 1a shows two additional folds occurring along this trajectory. In practice, when such a splitting occurs, the process is reinitialized by recomputing the eigendecomposition of $\mathbf{P}$ and selecting the eigenpair corresponding to the branch of interest.

As a second example, the inertia constant of the system's largest SG is gradually reduced from 50 to 5. Figure 1b illustrates the resulting trajectory of the system's least damped mode, initially located at $-0.4745 \pm j8.8572$ and corresponding to the local electromechanical mode of the SG at bus 31. Tracking based on (18) shows how the eigenvalue moves as inertia decreases, eventually crossing into the unstable region. In the same figure we verify that for the same step of $\Delta H = -0.5$, the use of TM for numerical integration outperforms FEM in terms of accuracy. Nevertheless, the

combined use of the numerical integration step with a corrector sub-routine tracks the trajectory with the prescribed tolerance.

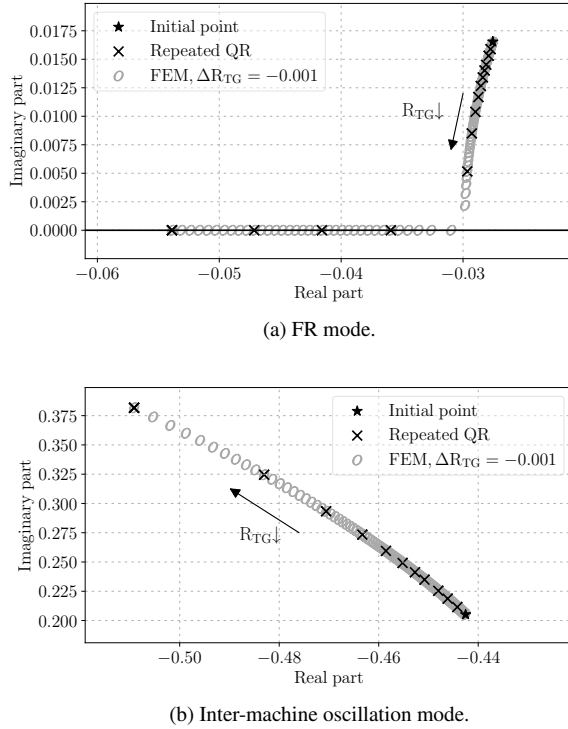
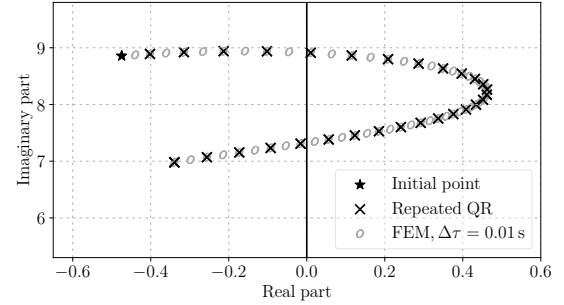


Figure 2: 39-bus system: tracking in the presence of multiple delays.

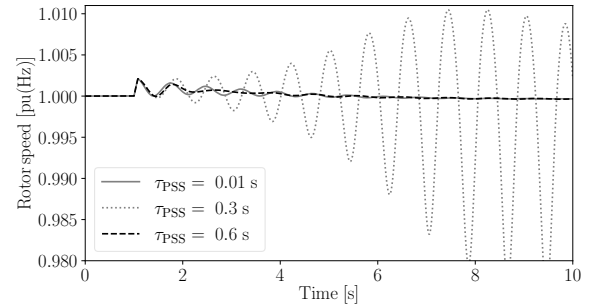
We next assess the method in the presence of multiple delays. To this end, we introduce delays to the input signals of the system's automatic voltage regulators (AVRs), PSSs and DER frequency and voltage controllers, yielding a total of 20 time-delayed variables. The resulting $\mathbf{P}$ has dimension 558×558 . For each delay τ_j , the spectral discretization in [14] uses N Chebyshev nodes in the interval $[-\tau_j, 0]$, which increases the size of $\mathbf{P}$ by $N + 1$. Thus, considering $N = 10$, the dimension of the approximated linear eigenvalue problem grows by 220 compared to the delay-free model. We then trace how variations in the SG droop constants affect the eigenvalues corresponding to (i) the system's FR mode and (ii) the inter-machine mode with the largest participation in the rotor speeds of the SGs at buses 32 and 39, initially located at $-0.4425 \pm j0.2052$. The calculated trajectories, shown in Fig. 2, confirm that the proposed tracking method accurately captures eigenvalue trajectories in the presence of multiple delays.

We next examine the tracking accuracy when the delay magnitude is treated as the continuation parameter, as described in Section 3.3. The analysis focuses on the local electromechanical mode of the SG at bus 31. The delay τ_{PSS} in the input signal of the corresponding PSS is varied from 0.01 to 0.6 s, producing the trajectory shown in Fig. 3a. As τ_{PSS} increases, the traced eigenvalue becomes progressively less damped and eventually crosses into instability for

$\tau_{PSS} > 0.1$ s. Further increasing τ_{PSS} beyond 0.28 s reverses this trend and eventually leads the system to regain small-signal stability for $\tau_{PSS} > 0.5$ s. This behavior is further validated via a time-domain simulation, considering a three-phase fault at bus 6, cleared after 80 ms by opening line 5–6. The response of the corresponding SG rotor speed is shown in Fig. 3b and confirms the conclusions drawn by the eigenvalue tracking. In the case of $\tau_{PSS} = 0.3$ s, the PSS's output introduces a destabilizing negative damping component to the SG's electrical torque, causing the rotor speed to diverge.



(a) Tracing a local electromechanical mode as delay is increased.



(b) Time-domain simulation for different delay values.

Figure 3: 39-bus system, SG at bus 31: PSS with input signal delay.

To study the effect of stochasticity, we assume the case in which a single delay is introduced in the frequency signal of the PSS connected to the SG at bus 31. We model the latency as a WAMS delay, with $\tau_0 = 0.01$ s, $p_{dr} = 10\%$, $\alpha = 0.01$ and $b = 2$. Starting from a very small value of $T = 0.001$ s, we increase the normal delivery period up to $T = 0.04$ s. The result is shown in Fig. 4. Alongside the above trajectory, we show the effect of increasing the magnitude of the delay when the latter is constant. We observe that for small values of T , the two variations have a similar effect, which is expected, since increasing the delivery period effectively increases the latency.

4.2. Irish System

We assess the scalability of the proposed eigenvalue tracking method using a 1,502-bus dynamic model of the

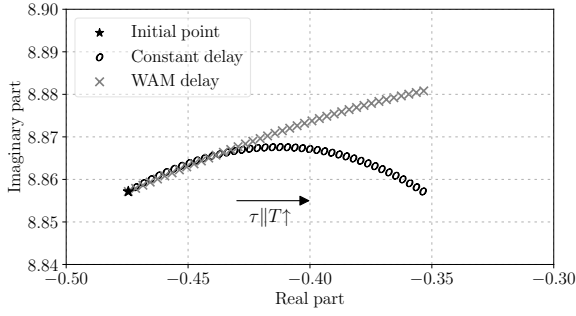


Figure 4: 39-bus system: effect of stochasticity on the PSS input signal delay.

All-Island Irish Transmission System (AIITS). The delay-free DAE model has 1,629 states and 9,897 algebraic variables. Its dimension, $11,526 \times 11,526$, renders the use of standard, dense QZ-based eigensolvers impractical. We first examine the accuracy of (18) in tracing poorly damped electromechanical modes of the AIITS. The DDAE model includes 28 delayed variables, affecting voltage and frequency measurements of the system's AVRs and PSSs. The delays, ranging from 0.009 to 0.022 s, increase the dimension of the approximated linear eigenvalue problem to $11,834 \times 11,834$. The effect of raising the gains of the system's PSSs from 1.5 to 10, is shown in Fig. 5. Although the examined eigenvalues are tightly clustered in the complex plane, the proposed approach efficiently traces all of them.

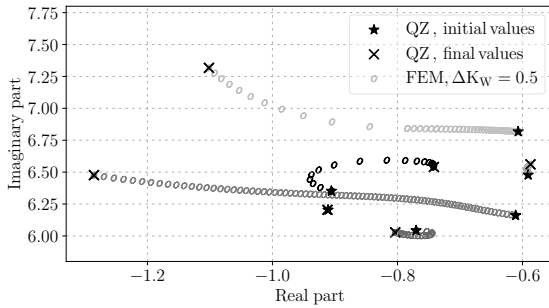


Figure 5: AIITS: tracking poorly damped electromechanical modes in the presence of multiple delays.

We next apply the formulation in Section 3.3 to trace the trajectories of the system's poorly damped electromechanical modes, when the time delay of the frequency signal of the PSS, at bus 717 increases from 0.01 to 0.5 s. The results, displayed in Fig. 6, confirm the accuracy of the proposed approach.

Lastly, we assess the computational efficiency of the proposed tracking approach. To this end we compare it with two eigendecomposition methods; QZ and KS, which are intended for dense and sparse matrices respectively. Results are summarized in Table 1.

²Projected, based on the total number of iterations needed, i.e., 49.

³The number of eigenvalues computed varies from 528 to 780.

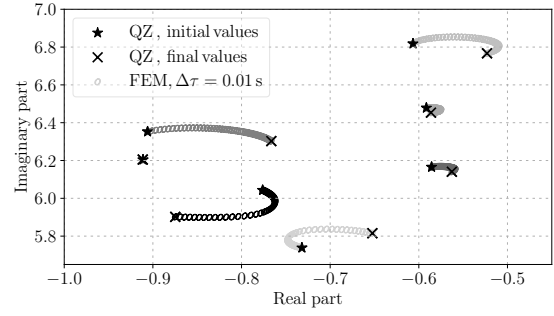


Figure 6: AIITS, SG at bus 717: increasing PSS input signal delay.

Table 1

AIITS: Tracking computational cost comparison for the case described in Fig. 6

	QZ	KS	Proposed Method (TM)	Proposed Method (FEM)
Single Step	3580 s	3.67 s	0.28 s	0.23 s
Total Time	175420 s ²	179.78 s ³	13.71 s	11.04 s

Both QZ and KS compute multiple eigenvalues at once. QZ computes the full spectrum of the discretized problem, whereas KS calculates only a selected part of it. However, repeated QZ and KS return unordered eigenpairs at each parameter value. Therefore, reconstructing eigenvalue trajectories requires an additional mode-matching step, which becomes nontrivial for large sparse systems with clustered modes or partial-spectrum computations. The total time reported for QZ and KS therefore accounts only for the eigensolution step, before any mode-matching procedure needed to connect successive eigenpairs. In addition, the accuracy, as well as the number of eigenvalues found by KS, highly depends on the tuning of the spectral transformation used. In our case, the *shift-and-invert* transformation is used. The computational time of KS can be reduced by targeting a smaller portion of the spectrum; however, selecting the spectral transformation and target region is nontrivial and typically requires trial and error, without guaranteeing a prescribed tolerance or the identification of the desired eigenvalue.

The proposed approach determines the trajectory of the targeted eigenpair in an automated manner and in a small fraction of the time required by the other methods. In the case considered in Table 1, FEM outperforms the TM. However, most of the computational time is used by the corrector sub-routine, which is set to an error tolerance of 10^{-8} . Based on the settings of the corrector, using the TM may lead to faster or more precise results, as it is implicit and unconditionally stable.

5. Conclusion

This paper proposes a continuation-based eigenvalue tracking technique for power systems impacted by time delays. The formulation retains the sparsity of the system's DDAE model and considers a continuation parameter which enables multiple system properties – including delay magnitudes – to be expressed as functions of it and varied simultaneously. Case studies demonstrate the accuracy and computational efficiency of the proposed approach. Future work will focus on applying the proposed approach to guide the design of delay-robust control schemes for DERs.

A. Appendix

In this section, we derive the expressions of $\mathbf{M}$ and $\mathbf{h}$ for the cases considered in Sections 3.2, 3.3, 3.4, i.e., for systems with multiple delays, systems including a varying delay and systems including a WAMS delay.

A.1. Systems with Multiple Delays

We define the functions $h_r(t) = e^{-s_r t} \cos(s_i t)$ and $h_i(t) = e^{-s_r t} \sin(s_i t)$ and consider the notation $h_r^j = h_r(\tau_j)$ and $h_i^j = h_i(\tau_j)$. By splitting real and imaginary parts of (20) and setting $\mathbf{C} = [\mathbf{A}_j h_r^j]^\top$, $\mathbf{S} = [\mathbf{A}_j h_i^j]^\top$, $\mathbf{C}_D = [\mathbf{A}'_j h_r^j]^\top$, $\mathbf{S}_D = [\mathbf{A}'_j h_i^j]^\top$, we obtain a system of the form (18), where

$$\mathbf{h} = \begin{bmatrix} (-s_r \mathbf{E}' + \mathbf{A}' + \mathbf{J}_\mu \mathbf{C}_D) \boldsymbol{\phi}_r + (s_i \mathbf{E}' + \mathbf{J}_\mu \mathbf{S}_D) \boldsymbol{\phi}_i \\ (-s_i \mathbf{E}' - \mathbf{J}_\mu \mathbf{S}_D) \boldsymbol{\phi}_r + (-s_r \mathbf{E}' + \mathbf{A}' + \mathbf{J}_\mu \mathbf{C}_D) \boldsymbol{\phi}_i \\ 0 \\ 0 \end{bmatrix}$$

$\mathbf{M}_3$ is the same as in (18) and:

$$\mathbf{M}_1 = \begin{bmatrix} s_r \mathbf{E} - \mathbf{A}_0 - \mathbf{J}_\mu \mathbf{C} & -s_i \mathbf{E} - \mathbf{J}_\mu \mathbf{S} \\ s_i \mathbf{E} + \mathbf{J}_\mu \mathbf{S} & s_r \mathbf{E} - \mathbf{A}_0 - \mathbf{J}_\mu \mathbf{C} \end{bmatrix}$$

$$\mathbf{M}_2 = \begin{bmatrix} \mathbf{E} \boldsymbol{\phi}_r + \mathbf{J}_{\tau\mu} (\mathbf{C} \boldsymbol{\phi}_r + \mathbf{S} \boldsymbol{\phi}_i) & -\mathbf{E} \boldsymbol{\phi}_i - \mathbf{J}_{\tau\mu} (\mathbf{C} \boldsymbol{\phi}_r - \mathbf{S} \boldsymbol{\phi}_i) \\ \mathbf{E} \boldsymbol{\phi}_i + \mathbf{J}_{\tau\mu} (\mathbf{C} \boldsymbol{\phi}_r - \mathbf{S} \boldsymbol{\phi}_i) & \mathbf{E} \boldsymbol{\phi}_r + \mathbf{J}_{\tau\mu} (\mathbf{C} \boldsymbol{\phi}_r + \mathbf{S} \boldsymbol{\phi}_i) \end{bmatrix}$$

with $\mathbf{J}_\mu = [\mathbf{I}_r \ \mathbf{I}_r \ \dots \ \mathbf{I}_r] \in \mathbb{R}^{r \times \mu r}$ and $\mathbf{J}_{\tau\mu} = [\tau_1 \mathbf{I}_r \ \tau_2 \mathbf{I}_r \ \dots \ \tau_\mu \mathbf{I}_r] \in \mathbb{R}^{r \times \mu r}$; $\mathbf{I}_r$ is the $r \times r$ identity matrix.

A.2. Systems with Delay as Varying Parameter

Splitting real and imaginary parts of (22) and by considering the functions h_r and h_i defined above for $\tau_\ell = p$, we arrive at the form of (18), where in this case:

$$\mathbf{h} = \begin{bmatrix} \mathbf{h}_1 \boldsymbol{\phi}_r + \mathbf{h}_2 \boldsymbol{\phi}_i \\ -\mathbf{h}_2 \boldsymbol{\phi}_r + \mathbf{h}_1 \boldsymbol{\phi}_i \\ 0 \\ 0 \end{bmatrix} \quad (33)$$

with

$$\mathbf{h}_1 = -\mathbf{A}_\ell h_r^\ell s_r - \mathbf{A}_\ell h_i^\ell s_i$$

$$\mathbf{h}_2 = \mathbf{A}_\ell h_r^\ell s_i - \mathbf{A}_\ell h_i^\ell s_r$$

$\mathbf{M}_3$ is the same as in (18) and:

$$\mathbf{M}_1 = \begin{bmatrix} s_r \mathbf{E} - \mathbf{A}_0 - \mathbf{A}_\ell h_r^\ell & -s_i \mathbf{E} - \mathbf{A}_\ell h_i^\ell \\ s_i \mathbf{E} + \mathbf{A}_\ell h_i^\ell & s_r \mathbf{E} - \mathbf{A}_0 - \mathbf{A}_\ell h_r^\ell \end{bmatrix}$$

$$\mathbf{M}_2 = \begin{bmatrix} \mathbf{E} \boldsymbol{\phi}_r + p \mathbf{A}_\ell (\boldsymbol{\phi}_r h_r^\ell + \boldsymbol{\phi}_i h_i^\ell) & -\mathbf{E} \boldsymbol{\phi}_i + p \mathbf{A}_\ell (\boldsymbol{\phi}_r h_i^\ell - \boldsymbol{\phi}_i h_r^\ell) \\ \mathbf{E} \boldsymbol{\phi}_i - p \mathbf{A}_\ell (\boldsymbol{\phi}_r h_i^\ell - \boldsymbol{\phi}_i h_r^\ell) & \mathbf{E} \boldsymbol{\phi}_r + p \mathbf{A}_\ell (\boldsymbol{\phi}_r h_r^\ell + \boldsymbol{\phi}_i h_i^\ell) \end{bmatrix}$$

If the system includes multiple delays, then we also consider the matrices $\mathbf{C}$, $\mathbf{S}$, $\mathbf{J}_\mu$ and $\mathbf{J}_{\tau\mu}$. Splitting real and imaginary parts of (25) we arrive at the form of (18), with $\mathbf{h}$ given by (33) and where in this case

$$\mathbf{h}_1 = -\mathbf{A}_\ell h_r^\ell s_r - \mathbf{A}_\ell h_i^\ell s_i$$

$$\mathbf{h}_2 = \mathbf{A}_\ell h_r^\ell s_i - \mathbf{A}_\ell h_i^\ell s_r$$

$\mathbf{M}_3$ is the same as in (18) and:

$$\mathbf{M}_1 = \begin{bmatrix} s_r \mathbf{E} - \mathbf{A}_0 - \mathbf{A}_\ell h_r^\ell - \mathbf{J}_\mu \mathbf{C} & -s_i \mathbf{E} - \mathbf{A}_\ell h_i^\ell - \mathbf{J}_\mu \mathbf{S} \\ s_i \mathbf{E} + \mathbf{A}_\ell h_i^\ell + \mathbf{J}_\mu \mathbf{S} & s_r \mathbf{E} - \mathbf{A}_0 - \mathbf{A}_\ell h_r^\ell - \mathbf{J}_\mu \mathbf{C} \end{bmatrix}$$

$$\mathbf{M}_2 = \begin{bmatrix} \mathbf{E} \boldsymbol{\phi}_r + \mathbf{J}_{\tau\mu} (\mathbf{C} \boldsymbol{\phi}_r + \mathbf{S} \boldsymbol{\phi}_i) + \mathbf{A}_\ell p (\boldsymbol{\phi}_r h_r^\ell + \boldsymbol{\phi}_i h_i^\ell) & \dots \\ \mathbf{E} \boldsymbol{\phi}_i + \mathbf{J}_{\tau\mu} (\mathbf{C} \boldsymbol{\phi}_r - \mathbf{S} \boldsymbol{\phi}_i) - \mathbf{A}_\ell p (\boldsymbol{\phi}_r h_i^\ell - \boldsymbol{\phi}_i h_r^\ell) & \dots \\ \dots & -\mathbf{E} \boldsymbol{\phi}_i - \mathbf{J}_{\tau\mu} (\mathbf{C} \boldsymbol{\phi}_r - \mathbf{S} \boldsymbol{\phi}_i) + \mathbf{A}_\ell p (\boldsymbol{\phi}_r h_i^\ell - \boldsymbol{\phi}_i h_r^\ell) \\ \dots & \mathbf{E} \boldsymbol{\phi}_r + \mathbf{J}_{\tau\mu} (\mathbf{C} \boldsymbol{\phi}_r + \mathbf{S} \boldsymbol{\phi}_i) + \mathbf{A}_\ell p (\boldsymbol{\phi}_r h_r^\ell + \boldsymbol{\phi}_i h_i^\ell) \end{bmatrix}$$

A.3. Communication Delays with Noise and Data Dropouts

We set $S_T^r = \Re\{S_T\}$, $S_T^i = \Im\{S_T\}$, $\tilde{S}_{TD}^r = \Re\{S_{TD}\} \mathbf{A}_1$ and $\tilde{S}_{TD}^i = \Im\{S_{TD}\} \mathbf{A}_1$. Splitting real and imaginary parts of (31) leads to a system in the form of (18), where in this case $\mathbf{M}_3$ is the same as in (18) and:

$$\mathbf{h} = \begin{bmatrix} (-s_r \mathbf{E}' + \mathbf{A}'_0 + S_T^r \mathbf{A}'_1) \boldsymbol{\phi}_r + (s_i \mathbf{E}' - S_T^i \mathbf{A}'_1) \boldsymbol{\phi}_i \\ -(s_i \mathbf{E}' - S_T^i \mathbf{A}'_1) \boldsymbol{\phi}_r + (-s_r \mathbf{E}' + \mathbf{A}'_0 + S_T^r \mathbf{A}'_1) \boldsymbol{\phi}_i \\ 0 \\ 0 \end{bmatrix}$$

$$\mathbf{M}_1 = \begin{bmatrix} s_r \mathbf{E} - \mathbf{A}_0 - S_T^r \mathbf{A}_1 & -s_i \mathbf{E} + S_T^i \mathbf{A}_1 \\ s_i \mathbf{E} - S_T^i \mathbf{A}_1 & s_r \mathbf{E} - \mathbf{A}_0 - S_T^r \mathbf{A}_1 \end{bmatrix}$$

$$\mathbf{M}_2 = \begin{bmatrix} (\mathbf{E} - \tilde{S}_{TD}^r) \boldsymbol{\phi}_r + \tilde{S}_{TD}^i \boldsymbol{\phi}_i & -(\mathbf{E} - \tilde{S}_{TD}^r) \boldsymbol{\phi}_i + \tilde{S}_{TD}^i \boldsymbol{\phi}_r \\ (\mathbf{E} - \tilde{S}_{TD}^i) \boldsymbol{\phi}_i - \tilde{S}_{TD}^r \boldsymbol{\phi}_r & (\mathbf{E} - \tilde{S}_{TD}^i) \boldsymbol{\phi}_r + \tilde{S}_{TD}^r \boldsymbol{\phi}_i \end{bmatrix}$$

CRedit authorship contribution statement

A. Bouterakos: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **G. Tzounas:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

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